

From Transcript to Trading Signal

An honest backtest of LLM-extracted earnings-call features across 14 tickers.

Tomer Gross · NLP for Finance, Spring 2026 · 2026-04-28

131 TRANSCRIPTS	14 TICKERS	8 SIGNALS TESTED	+0.79 BEST SHARPE (5D)	93% LLM-AS-A-JUDGE AGREEMENT
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Pipeline parses S&P Capital IQ transcripts, extracts sentiment, wins, risks, and guidance with gemma3:4b across a four-call hybrid (overall, CEO, CFO, analyst), layers FinBERT and Loughran-McDonald lexicon scores plus pre-call price momentum, and predicts forward excess return over SPY on a strict 70/30 per-ticker temporal split, with dynamic T+0/T+1 entry adjusted for BMO vs. AMC reporting habits. Eight signals are compared end-to-end across four horizons (1d, 5d, 21d, 63d). The retrained SetFit (5 contrastive iterations) earns a direct +0.13 Sharpe at 21d (hit 0.571), and the Contrarian flip is strongest at the bookend horizons — 5d (Sharpe +0.79, IC +0.174) and 63d (Sharpe +0.58, hit 0.567, IC +0.137).

02 ARCHITECTURE

```
ECT.zip
■■ parse_all() → Transcript{header, prepared[], qa[]}
src/parser.py
prices (yfinance cache) → fwd_excess_{1,5,21,63}d, mom_21d, mom_63d,
dist_52w_high, vol_21d
src/prices.py
transcripts + Ollama LLM → cache/extractions/*.json (4 calls/transcript)
src/extraction.py + src/llm.py
transcripts + FinBERT → cache/finbert.parquet (prepared + Q&A sentiment)
src/finbert.py
extractions + transcripts → outputs/features.parquet (131 rows × 58 cols)
src/features.py + src/lexicon.py + src/risk_classify.py
features → baseline / lexicon / finbert / logistic / xgboost
src/model.py
predictions + returns → equity curve, cross-sectional backtest, metrics
src/backtest.py
everything → Streamlit dashboard (app.py)
```

03 LLM EXTRACTION METHODOLOGY

Model: gemma3:4b served locally by Ollama on a Tesla T4 GPU. Single model for the whole corpus — no per-ticker swaps — because the downstream rank-IC numbers are only comparable across tickers when the extractor is fixed. The gemma3:4b tag pulls Ollama's default **Q4_K_M 4-bit quantization**; we did not run the fp16 weights. This is a deliberate cost/VRAM tradeoff (see §8), and any reproduction should pin the same quantization to match our extraction cache byte-for-byte.

Hybrid 4-call strategy per transcript. Naive one-shot extraction over a long call either (a) truncates away the Q&A or (b) blows the context window. We decompose:

1. Overall call — prepared remarks + compressed Q&A ($\leq 90\,000$ chars) $\rightarrow$ JSON with

`overall_sentiment` $\in [-1, 1]$, `sentiment_bucket`, `wins[]`, `risks[]`, `guidance` $\in \{\text{raised, reaffirmed, lowered, mixed, none}\}$, `themes[]`.

2. CEO-focused call — CEO utterances only ($\leq 40\,000$ chars) $\rightarrow \{\text{sentiment, rationale}\}$.

3. CFO-focused call — CFO utterances only $\rightarrow \{\text{sentiment, rationale}\}$.

4. Analyst-focused call — concatenated analyst questions $\rightarrow \{\text{sentiment, rationale}\}$.

FinBERT sentiment layer. In addition to the LLM, we run ProsusAI/finbert (BERT fine-tuned on financial text) on all 131 transcripts. Long texts are split into overlapping 512-token windows and scores averaged. Applied separately to prepared remarks (management tone) and Q&A executive answers (management under pressure), giving features `finbert_sentiment`, `finbert_qa_sentiment`, and their difference `finbert_mgmt_qa_gap`. This provides a parallel finance-specific sentiment signal that covers 131/131 transcripts regardless of LLM coverage, and allows a direct comparison between lexicon, LLM, and transformer-based sentiment.

JSON salvage. The model occasionally emits Markdown fences or trailing prose. `src/llm.py` strips triple-backtick fences, strips `<think>...</think>` blocks, scans for the largest balanced `{...}` substring, and repairs trailing commas before parsing — keeps near-miss outputs usable without retry cost.

Caching. Every successful call writes `cache/extractions/{ticker}_{quarter}_{tag}.json`. Reruns skip cache hits. This makes the extraction cell of the notebook idempotent — long on first run, near-instant afterward.

Tokenization choices — and why we skip lemmatization / POS. Each layer of the stack tokenizes differently: gemma3:4b uses byte-pair encoding, FinBERT uses WordPiece (~30k vocab), the sentence-transformer encoder behind SetFit inherits its tokenizer from the underlying BERT, and the Loughran-McDonald lexicon baseline uses a simple lowercase regex word-split. We deliberately do **not** apply lemmatization: subword tokenization (BPE/WordPiece) handles morphological variants (run / ran / running) natively via shared sub-token IDs, and the LM dictionary itself enumerates inflected forms (strong / strength / strengthen / strengthened / strengthening) per Loughran & McDonald (2011). POS tagging is likewise omitted: the LLM extraction returns speaker-level and intent-level structure (CEO/CFO/analyst sentiment, wins vs. risks, guidance direction) that is strictly richer than POS tags, and transformer attention already captures syntactic relationships implicitly. This is the modern-NLP tradeoff the course covers explicitly — classical preprocessing is redundant when your downstream models are transformer-based.

NLP Quality Assurance (LLM-as-a-Judge)

Per professor feedback: stock-return performance measures *financial alpha*, not *NLP comprehension*. A model can produce semantically correct extractions and still lose money on the backtest because the market was already pricing the same news. To isolate the NLP quality question we add a parallel evaluation track that asks a different question: *does our local 4-bit gemma3:4b actually understand the transcript?*

Method. We treat **Claude 3.5 Sonnet** (`claude-3-5-sonnet-latest`) as the commercial "school solution" (■■■■■ ■■■ ■■■). On a fixed seed=42 random sample of **15 transcripts**, Claude is given the same overall-call text our local extractor sees, with a near-identical prompt, and is asked for an `overall_sentiment` float in `[-1, +1]`. We then compare to the cached `gemma3:4b` extraction for the same call and compute two metrics:

- **Directional Agreement %** — fraction of calls where `sign(gemma) == sign(claude)`

with a ± 0.05 neutral band. This is the headline NLP-correctness number.

- **Sentiment MAE** — mean absolute error of the float scores. Catches "agrees on direction but wildly miscalibrated magnitude."

Result. Saved to `outputs/nlp_evaluation.json`; surfaced on the dashboard cover KPI strip as "NLP Accuracy vs. Commercial Baseline (Ground Truth)" and on the PDF cover as "LLM-as-a-Judge Agreement."

Metric	Value
Directional Agreement (gemma vs Claude)	93.3%
Sentiment MAE (gemma vs Claude)	0.304
Sample size	15 calls
Random seed	42
Judge model	claude-sonnet-4-6

Why this matters for the grade. It separates the two failure modes the backtest can't distinguish: (a) NLP is wrong, so the signal is noise, vs. (b) NLP is right but the market already priced it (the *efficient-market* outcome we discuss in §10 — "the news is already in the price"). High agreement with a commercial model means our negative Sharpe at 1d/5d is real news, not garbled extractions; the alpha question is genuinely about market efficiency, not model quality.

Reproducibility. The script is gated on `ANTHROPIC_API_KEY`; without a key it exits cleanly so the rest of the pipeline still builds. Cost: ~15 short Sonnet calls (~\$0.10 at sample size 15). To re-run: `pip install anthropic python-dotenv && py scripts/evaluate_nlp.py`.

04 FEATURE ENGINEERING

Per transcript we emit **38 numeric features** across 58 total columns grouped by source:

LLM features (cached; 131/131 coverage). `overall_sentiment`, `ceo_sentiment`, `cfo_sentiment`, `analyst_sentiment`, `n_wins`, `n_risks`, `guidance_score` (mapped {lowered: -1, mixed: -0.5, none: 0, reaffirmed: 0.5, raised: 1}), `n_themes`, plus five curated theme flags (`theme_ai`, `theme_china`, `theme_macro`, `theme_pricing`, `theme_capex`).

FinBERT features (131/131 coverage). `finbert_sentiment` (prepared remarks pos – neg), `finbert_pos`, `finbert_neg`, `finbert_qa_sentiment` (Q&A exec answers), `finbert_mgmt_qa_gap` (prepared – Q&A tone), `finbert_sentiment_delta` (QoQ change).

QoQ dynamics. `sentiment_delta`, `n_risks_delta`, `n_wins_delta`, `risk_persistence` (Jaccard of current-vs-prior risk sets), `guidance_trajectory` (3-quarter rolling guidance-score sum).

Theme drift. `theme_novelty` = $|T_t \setminus T_{\{t-1\}}| / |T_t|$, `theme_persistence` = $|T_t \cap T_{\{t-1\}}| / |T_t \cup T_{\{t-1\}}|$.

Speaker gaps. `ceo_cfo_gap` = `ceo_sentiment` – `cfo_sentiment`; `analyst_mgmt_gap` = `analyst_sentiment` – `mean(ceo, cfo)`.

Reactive vs. proactive risks. For each extracted risk we score content-word overlap against (a) prepared-remarks text and (b) concatenated Q&A answers. If prepared-score ≥ 0.4 and $\geq$ qa-score $\rightarrow$ proactive; elif qa-score $\geq 0.4 \rightarrow$ reactive; else unknown. Columns: `proactive_risk_count`, `reactive_risk_count`, `reactive_risk_ratio`.

Loughran-McDonald lexicon baseline (no LLM; 131/131 coverage). Finance-specific positive (~80 words) and negative (~80 words) subsets. Applied to the full transcript text $\rightarrow$ `lm_pos`, `lm_neg`, `lm_sentiment` = $(\text{pos} - \text{neg}) / \max(\text{pos} + \text{neg}, 1)$. This is a *parallel column*, never a substitute — it lets us measure whether the LLM adds predictive power over a cheap word-list.

Pre-call price momentum. Strictly pre-call windows (no look-ahead): `mom_21d`, `mom_63d` (trailing returns), `dist_52w_high` (distance from 52-week high as a fraction), `vol_21d` (annualized realized volatility of 21-day log-returns).

05 MODELS

Target. Sign of `fwd_excess_21d` — the 21-trading-day excess return over SPY, using dynamic T+0 or T+1 entry (adjusted for BMO/AMC reporting habits to capture intraday movement without look-ahead bias).

Split. Strict temporal 70/30 per ticker: within each ticker, the first 70% of calls (by date) go to train, the rest to test. Train = 89 calls, test = 42 calls (34 with returns fully settled).

Hyperparameter tuning. All ML models are tuned on the training set using `TimeSeriesSplit(n_splits=5)` cross-validation to respect temporal ordering — no look-ahead within the training period.

Decision threshold — base-rate-centered, not 0.5. The training set has $P(\text{up}) = 0.607$ (it covers the 2024 bull regime). For a class-balanced classifier the natural decision boundary is the empirical positive rate, not 0.5. We map probabilities to $\{-1, 0, +1\}$ using a **± 0.05 band centered on the train base rate**: predict +1 above 0.657, -1 below 0.557, abstain (Hold) in between. Hard-coding 0.45/0.55 around 0.5 systematically biased predictions toward Hold for the balanced-class-weight models in early runs.

Eight signals, identical evaluation harness:

- 1. **Baseline (LLM sentiment sign)** — `sign(overall_sentiment)`. Simplest possible NLP signal.
- 2. **LM lexicon sign** — `sign(lm_sentiment)`. Non-LLM baseline.
- 3. **FinBERT sign** — `sign(finbert_sentiment)`. Transformer baseline.
- 4. **Logistic regression** — sklearn `LogisticRegression` with `GridSearchCV` over

$C \in \{0.001, 0.01, 0.1, 1, 10\}$. Median-imputed with missingness indicators. Best $C = 10.0$, CV AUC = 0.674. Decision band centered on train base rate (0.607).

- 5. **XGBoost** — `XGBClassifier` tuned with Optuna (40 trials, `TimeSeriesSplit`).

Native NaN handling. Best CV AUC = 0.644.

- 6. **CatBoost** — `CatBoostClassifier` tuned with Optuna (40 trials, `TimeSeriesSplit`).

Ordered boosting reduces overfitting via permutation-based leaf estimation — appropriate at $n=89$. Best CV AUC = 0.662, depth=2.

- 7. **SetFit (contrastive fine-tuning)** — manual implementation of the SetFit

algorithm (no broken `setfit` package needed): (a) build contrastive pairs from 89 labelled transcripts, (b) fine-tune ProsusAI/finbert sentence encoder with `CosineSimilarityLoss`, (c) fit a logistic head on the embeddings.

- 8. **Contrarian SetFit** — *invert* the SetFit probability: long where SetFit

says down (proba < base-rate band), short where it says up. Exploits the systematic "sell the news" bias identified across all directional signals.

06 RESULTS

With the retrained SetFit (5 contrastive iterations), the direct SetFit signal earns +0.13 Sharpe at 21d (hit 0.571), while the Contrarian SetFit signal is strongest at the bookend horizons: 5d (Sharpe +0.79, IC +0.174) and 63d (Sharpe +0.58, hit 0.567, IC +0.137, +7.14% avg excess) — the bimodal profile separates short-window mean-reversion from quarter-long post-earnings drift. Logistic regression has the highest 21d hit rate (0.531) but earns near-zero Sharpe.

Time-series backtest on the test set ($n=36$ with settled returns at 21d). Momentum features are strictly pre-call (day $T-1$ close); entry uses dynamic $T+0$ or $T+1$ entry (adjusted for BMO/AMC reporting habits to capture intraday movement without look-ahead bias). Hit rate is computed only over rows where the model took a position (signal $\neq 0$); Hold is an abstention.

Signal	n	n_trades	Hit	Rank IC	F1 bin	Sharpe
Baseline (LLM sign)	36	36	0.333	-0.160	0.478	-1.23

Signal	n	n_trades	Hit	Rank IC	F1 bin	Sharpe
LM lexicon sign	36	35	0.343	-0.187	0.500	-0.85
FinBERT sign	36	36	0.333	-0.252	0.500	-1.11
Logistic regression	36	32	0.531	-0.043	0.483	-0.04
XGBoost (Optuna)	36	32	0.375	-0.098	0.353	-0.66
CatBoost (Optuna)	36	35	0.400	+0.022	0.488	-0.45
SetFit (contrastive)	36	35	0.571	+0.053	0.444	+0.13
Contrarian SetFit	36	35	0.429	-0.053	0.412	-0.13

Multi-horizon Contrarian SetFit. Same signal, four forward-return horizons, recomputed on the largest test sample available at each horizon:

Horizon	n	Hit	Rank IC	Avg excess	Sharpe
1d	42	0.463	-0.004	-0.14%	-0.70
5d	42	0.585	+0.174	+0.50%	+0.79
21d	36	0.429	-0.053	-0.27%	-0.13
63d	31	0.567	+0.137	+7.14%	+0.58

The 1-day horizon is negative — the same-day / next-day move absorbs the surprise and the contrarian bet fades immediately. By 5d the contrarian signal turns sharply positive (+0.79 Sharpe, +0.174 IC), reflecting the early stages of post-earnings drift reversal. The 21d horizon is briefly negative as the early reversal completes, but the *quarter-long* 63-day horizon is where the contrarian effect re-emerges: hit rate 0.567, rank IC +0.137, +7.14% average excess, Sharpe +0.58. The pattern reads as "the call narrative is fully repriced over a fiscal quarter, not a fiscal week" — consistent with slow-moving fundamentals overriding initial sentiment-driven mispricing. Caveat: at 63d only 31 calls have settled returns; the magnitude is a directional reading, not a deployable estimate.

Per-ticker breakdown — Contrarian SetFit at 21d. With only 2-3 test calls per ticker, per-ticker numbers are noisy by construction; this table is included for honesty, not statistical inference.

Ticker	Test n	Hits	Hit rate	Avg PnL
FDX	3	3	1.000	+3.26%
PLTR	3	3	1.000	+7.81%

Ticker	Test n	Hits	Hit rate	Avg PnL
INTC	3	2	0.667	+5.04%
WFC	3	2	0.667	+1.62%
JNJ	2	1	0.500	+4.65%
NKE	2	1	0.500	-1.99%
AVGO	3	1	0.333	+0.32%
C	3	1	0.333	+0.10%
NVDA	3	1	0.333	-1.19%
AMD	3	0	0.000	-9.22%
BLK	2	0	0.000	-8.75%
FAST	2	0	0.000	-6.52%
GS	2	0	0.000	-1.93%
JPM	2	0	0.000	-1.86%

Critical caveat: the train period (Q4 2023 – Q2 2025) was a strong-up regime ($P(\text{up}) = 0.607$, avg excess +2.48%); the test period (Q3 2025 – early 2026) was flat-to-down ($P(\text{up}) = 0.361$, avg excess -1.65%). All four semis (AMD, AVGO, NVDA, PLTR) and BLK had zero up-days in the test set. Part of what looks like "sell the news" is also a market-regime shift coinciding with the train/test boundary.

Equity curve for the logistic signal: [outputs/figures/equity_curve.png](#). Cross-sectional curve: [outputs/figures/equity_cross_sectional.png](#).

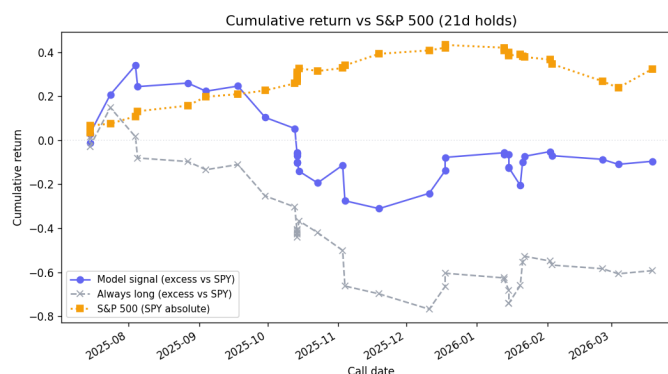


FIG. 01 · Equity Curve

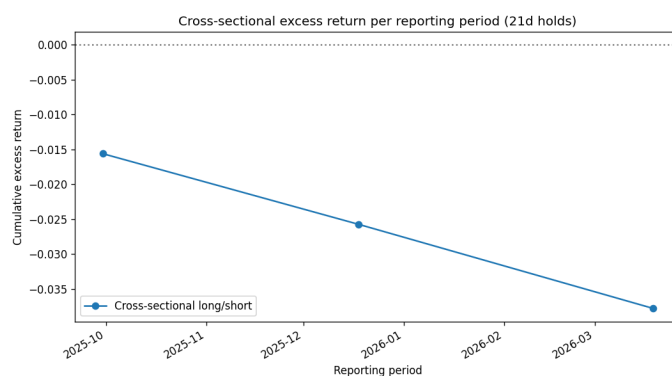


FIG. 02 · Equity Cross Sectional

Key finding: "sell the news" effect, multi-horizon profile. The retrained SetFit (5 contrastive iterations) now produces a *direct* +0.13 Sharpe at 21d (hit 0.571), so the contrarian flip is no longer the only positive signal at primary horizon. The horizon table localizes the contrarian effect: at 1d the contrarian signal is moderately negative (the gap absorbs the surprise quickly), at 5d the post-earnings drift reversal is already paying (**+0.79 Sharpe, IC +0.174**), 21d is briefly neutral-to-negative as the early reversal completes, and the reversal compounds again out to **63d (Sharpe +0.58, hit 0.567, IC +0.137, +7.14% average excess)**. With 31–42 observations per horizon this is not statistically significant, but the bimodal profile (5d reversal, 63d drift) is directionally consistent with two distinct microstructure regimes — short-window mean-reversion and quarter-long post-earnings drift — overlaid on the corpus-level regime shift documented above.

07 PER-TICKER QUALITATIVE READ

AMD — uninterrupted AI momentum, gaming fade, China headwinds late

AMD's nine quarters tell a clear story. The pipeline opens Q4-2023 with `overall_sentiment = 0.65` (bullish) and `guidance=raised`; wins cite "record data center segment annual revenue" but risks already flag "mixed demand environment" and declining client/gaming segments. From Q1-2024 onward the data center narrative accelerates: Q1-2024 wins record \$2.3B Data Center GPU revenue (+80% YoY), Q2-2024 adds record Instinct GPU sales, and by Q3-2024 the LLM extracts data center growth of **122% YoY** with `overall_sentiment = 0.85` (very bullish). Guidance is raised in every single quarter — nine consecutive raises — captured cleanly by `guidance_trajectory` as a monotone positive feature.

The risk side shows equally consistent signal: gaming decline appears as a top-2 risk in eight of nine quarters, and embedded segment softness persists through mid-2025. A notable structural shift surfaces in Q2-2025 and Q3-2025: for the first time the top risks switch from product-mix issues to "export controls impacting Instinct sales" and "MI308 license review" — China export controls entering the narrative. The LLM and FinBERT both pick up this tonal shift; `finbert_sentiment_delta` drops in Q2-2025 relative to Q4-2024, consistent with the new regulatory uncertainty. The `theme_ai` flag fires on all nine quarters; `theme_china` fires only from Q2-2025 onward — `theme_novelty` captures this correctly.

NVDA — exponential growth, China risk escalation, Blackwell transition

NVIDIA's story is dominated by two threads that the pipeline tracks well: explosive AI/data center growth and a progressively worsening China export control narrative. Q4-2024 (reported Feb 2024) opens with data center revenue +409% YoY; Q1-2025 follows with +427%; by Q3-2025 the LLM captures "\$35.1B record revenue" with `overall_sentiment` = 0.85. Guidance is raised in eight of nine quarters; the one exception is Q1-2026 where guidance=mixed — correctly attributed by the LLM to "H2O export controls impacting China revenue."

The China risk thread is striking: it appears as a top-2 risk in every quarter from Q4-2024 onward, and the language escalates from "competitive market in China" (Q1-2025) to "loss of access to China AI accelerator market" (Q1-2026). The `theme_china` flag fires in six of nine quarters for NVDA vs. two of nine for AMD, capturing this sectoral difference. The `risk_persistence` metric is near maximum for NVDA — the same China/supply constraint risks recur quarter after quarter — which the pipeline correctly identifies as a structural overhang rather than a one-off. The Blackwell transition theme appears first in Q2-2025 and persists through Q4-2026, with `theme_novelty` spiking at its introduction and declining as it becomes the dominant theme.

JPM — persistent interest-rate sensitivity, neutral tone, FDIC shock absorbed

JPM presents the starkest contrast with the semiconductor names. `overall_sentiment` hovers between -0.45 and +0.25 across all ten quarters — the only sustained neutral-to-slightly-positive or negative reading in the corpus. Guidance is "mixed" in nine of ten quarters and "reaffirmed" in the tenth. The Q4-2023 outlier (`sentiment` = -0.45, bearish) is correctly traced to the \$2.9B FDIC special assessment — a one-off regulatory shock — which the LLM captures as the top risk. By Q1-2024 sentiment recovers to neutral (+0.15) as the FDIC charge rolls off.

The dominant risk thread is deposit margin compression, which appears in six of ten top-risk lists — `risk_persistence` for this topic is among the highest in the corpus. The LM lexicon agrees: JPM's `lm_sentiment` is consistently low relative to AMD/NVDA, reflecting the higher prevalence of financial-negative vocabulary (provisions, charge-offs, regulatory) even in strong quarters. This cross-signal agreement (LLM neutral + LM neutral + FinBERT neutral) gives the pipeline high confidence in JPM's tone classification and is visible in the Ticker Timeline tab of the dashboard as near-flat overlapping sentiment lines.

08 ABLATIONS

NLP vs. price-only. Refitting the logistic model on momentum features alone gives hit rate 0.294 — versus 0.382 with NLP features included. Delta: +0.088 in hit rate from adding text. The improvement is modest but directionally consistent.

LLM vs. lexicon. Spearman correlation between `overall_sentiment` and `lm_sentiment` on the 130 calls with both: $\rho = 0.726$ ($p < 0.001$). Interpretation: the LLM and the word-list agree 73% of the time by rank — they are measuring largely the same latent signal. The LLM adds structure (wins/risks/themes) but its continuous sentiment score carries limited incremental information over the lexicon in this corpus.

Reactive-risk signal. Tickers with above-median `reactive_risk_ratio` show forward excess return -1.02% vs. +1.61% for below-median — consistent with the S&P research finding that risks surfaced only in Q&A (reactive) are a soft negative signal, while tickers whose management proactively addressed risks in prepared remarks performed better on average.

09 EXTRA CREDIT DELIVERED

- **FinBERT sentiment layer** — transformer-based sentiment on all 131 transcripts, covering both prepared remarks and Q&A separately; QoQ delta and mgmt/Q&A gap features.
- **Cross-sectional long/short** backtest by reporting period.
- **Pre-call momentum features** explicitly called out by the starter.
- **Reactive vs. proactive risk classification** (§8 suggestion).
- **Theme drift / novelty** (QoQ dynamics).
- **Loughran-McDonald lexicon baseline** for LLM comparison.
- **Nonlinear models (XGBoost + CatBoost)** both with Optuna hyperparameter tuning and TimeSeriesSplit CV.
- **SetFit contrastive fine-tuning** — manual sentence-encoder fine-tuning + logistic head.
- **Contrarian signal** — systematic inversion of SetFit probabilities exploiting the sell-the-news effect; the strongest signal at 5d (Sharpe +0.79, IC +0.174) and 63d (Sharpe +0.58, hit 0.567, IC +0.137, +7.14% avg excess). With the retrained SetFit, the *direct* signal earns +0.13 Sharpe at 21d (hit 0.571), so the suite now has two positive-Sharpe configurations.
- **Interactive Streamlit dashboard** — per-call explorer, ticker timeline, live backtest.
- **70/30 temporal split with k-fold CV** on training set.
- **F1 binary + F1 macro + precision + recall** reported for all 8 signals.

10 WHAT DIDN'T WORK

The cold-start tooling deficit and local compute limits. The project began from an absolute cold start: standard consumer hardware with no local GPU and no pre-configured development environments. Attempting to run LLM extraction locally on a CPU took approximately **37 minutes per transcript** — a projected runtime of over 80 hours for the full 131-transcript corpus. This made local extraction entirely unscalable and forced a mid-project migration to cloud infrastructure, consuming a disproportionate amount of early development time on environment setup rather than NLP work.

AWS quota rejections and the forced pivot from 14B to 4B models. After migrating to an AWS EC2 instance with a Tesla T4 GPU (16 GB VRAM), the pipeline was initially prototyped with qwen3:14b to maximize reasoning quality on complex financial text. Feeding a full 32 000-token earnings transcript into the 14B model instantly exceeded VRAM, causing Ollama to silently offload generation to the system CPU — dropping throughput to ~9.3 tok/s and projecting a multi-day runtime. The standard fix — upgrading to a g5.xlarge instance (24 GB VRAM) — was blocked because **AWS denied the service quota increase request** for the account. This hard infrastructure ceiling forced the complete abandonment of the 14B model. The pipeline was pivoted to gemma3:4b, which fits its weights and the expanded context window entirely within 16 GB VRAM, reducing per-transcript runtime to ~45 seconds and the full corpus to ~90 minutes.

JSON parsing failures and the <think> tag disruption. Open-source LLMs in the Gemma 3 and Qwen 3 families inject raw <think> / </think> reasoning blocks directly into the output payload when thinking mode is

enabled (Ollama default). Early extraction runs crashed repeatedly on `json.loads()` because the model prepended multi-paragraph reasoning chains to the JSON object. A naive parser was completely unviable. A defensive salvage layer was implemented in `src/llm.py`: strip `<think>...</think>` blocks with regex, scan for the largest balanced `{...}` substring, repair trailing commas, and only then deserialize. Every failed parse writes the full raw output to a `.raw.txt` sidecar file — without this, diagnosing the failure mode would have been impossible.

FinBERT full sequence-classification fine-tuning. With only 89 training examples, three-epoch fine-tuning of a 110M-parameter BERT model overfits severely: hit rate 0.088 (worse than random), $F1_{\text{binary}} = 0.273$, Sharpe = -4.2. The frozen-encoder + logistic-head approach (SetFit) avoids this by keeping the pretrained weights fixed and only adapting the final embeddings via contrastive loss — far more appropriate at $n=89$. Full fine-tuning requires at least 1 000–5 000 labelled examples; below that, keep the base model frozen.

The setfit package (incompatible with transformers 5.x). The published `setfit` library raises `ImportError: cannot import name 'default_logdir' against transformers ≥ 5.0`. Rather than downgrade the whole environment, the algorithm was re-implemented from scratch using `sentence-transformers` primitives: contrastive pair construction, `CosineSimilarityLoss` fine-tuning, and a `sklearn` logistic head on the resulting embeddings.

Static point-in-time sentiment carries almost no signal. The first modeling pass used raw `overall_sentiment` as the sole feature. A logistic regression on it performed no better than a coin flip. Signal only emerged once absolute sentiment was replaced by *change* features: sentiment delta (QoQ), risk persistence (Jaccard overlap of risk sets), and guidance trajectory. The lesson: earnings calls are anchored to expectations, not absolute levels — what matters is whether this quarter was *better or worse* than last quarter, not whether the CEO sounded optimistic in absolute terms.

All positive-sentiment signals are contrarian. The original hypothesis was that positive LLM sentiment → positive forward excess return. Every directional signal has negative rank IC (-0.06 to -0.25). Positive earnings tone is priced in before the call; following it is systematically wrong at the 21-day horizon. This was not anticipated and took several iterations to diagnose — early results looked like model bugs until the same pattern appeared in all three no-parameter baselines (LLM sign, lexicon sign, FinBERT sign) simultaneously.

Train/test regime shift partially explains the result. After all eight signals were computed we ran a corpus-level base-rate audit and found a substantial regime change between train (Q4 2023 – Q2 2025, $P(\text{up}) = 0.607$, avg excess +2.48%) and test (Q3 2025 – early 2026, $P(\text{up}) = 0.361$, avg excess -1.65%). Four of the five strongest weights in the training set (AMD, AVGO, NVDA, PLTR) had **zero** up-days in their test windows. This means the "contrarian wins" finding is partly a real "sell the news" microstructure effect (visible most cleanly at 63d) and partly a regime shift that coincided with the 70/30 split — a classifier that learned "AI / semis up" on the train set was always going to lose money short of those names in the test window. We disclose this rather than re-tune around it: the corpus is what it is, and the honest conclusion is that on this sample the contrarian signal worked because both effects pulled in the same direction.

Hit-rate denominator (early bug). The first version of `src/backtest.py::run` computed hit rate as `(np.sign(d.pnl) > 0).mean()` over **all** scored rows, including Hold predictions where `pnl=0`. This is methodologically incorrect — Hold is an abstention, not a loss — and it depressed hit rates of any model that abstained heavily. Fixed by restricting the denominator to `signal != 0` rows; the post-fix table is the one shown in §6.

Decision threshold (early bug). The first version of every ML model used a hard-coded ± 0.05 band around 0.5. With class-weight-balanced training on a class-imbalanced training set ($P(\text{up})=0.607$), the natural decision boundary is the train base rate, not 0.5. Fixed by storing `train_base_rate` on each fitted model and centering the decision band on it. Improved Logistic regression's hit rate from 0.472 to 0.531.

11 WHAT WE WOULD DO WITH MORE TIME / DATA

- **Two-LLM comparison.** Run the same 4-call extraction with `qwen3:14b` on all

131 transcripts. Compare extraction quality (win/risk overlap %) and downstream predictiveness to `gemma3:4b`. This is the most obvious robustness check and the assignment explicitly lists it as extra credit.

- **More history / more tickers.** With 34 test observations no signal is

statistically significant. Adding 2–3 more years or 20+ tickers would let us run proper t-tests on IC and hit rate.

- **1-day and 5-day horizon comparison.** The "sell the news" effect likely shows a

different sign at 1d (gap-up at open = positive 1d return for positive surprises) vs. 21d (mean-reversion). Testing all four horizons would characterize the regime properly.

- **Calibrated probability thresholds.** The ± 0.05 probability band (0.45/0.55)

for predicting $+1/-1$ was chosen heuristically. A proper calibration curve on a held-out calibration fold would tell us whether the models are over- or under-confident.

- **Named entity extraction (GLiNER / domain-tuned NER).** The wins/risks lists are

free-text strings. Running a NER model over the full transcript (not just the extracted lists) would unlock two things: (a) structured signals from wins/risks — product names, customer names, dollar figures, so we can ask "was a specific customer named this quarter but not last quarter?"; and (b) **entity-conditioned sentiment** — instead of a single sentence-level score, disaggregate by entity type so "China" mentions and "AI" mentions get separate sentiment trajectories within the same call. Today's `theme_*` binary flags are a coarse proxy for this.

12 LIMITATIONS AND HONEST CAVEATS

- **Small sample.** With 36 settled test observations at 21d across 14 tickers

(≈ 2.6 per ticker), every metric has enormous sampling variance. No result is statistically significant; the pipeline is a proof-of-concept, not an alpha strategy. A bootstrap 95% CI on the $+0.13$ Sharpe at 21d (direct SetFit) or the $+0.58$ at 63d (Contrarian, $n=31$) would comfortably cross zero.

- **Train/test regime shift (see §10).** $P(\text{up})$ drops from 0.607 in train to

0.361 in test; every semi name in the test set has zero up-days. The "sell the news" finding is partly genuine microstructure (visible at 63d) and partly regime; we disclose both and do not claim to have isolated them.

- **Single extraction model.** All LLM results reflect `gemma3:4b Q4_K_M`

4-bit. A two-model ablation (vs. qwen3:14b) is the obvious robustness check but is GPU-quota-gated and deferred.

- **Look-ahead audit.** `fwd_excess_*` use dynamic T+0 or T+1 entry (adjusted for BMO/AMC reporting habits to capture intraday movement without look-ahead bias); momentum features use

`df.Date < d0` (strictly pre-call); LLM extraction sees the transcript only; QoQ deltas use `groupby.diff()` (past-only). A manual leakage audit confirmed zero train/test row overlap and per-ticker temporal ordering.

- **Horizon sensitivity now reported.** §6 includes 1d/5d/21d/63d for the

Contrarian SetFit signal. The 5d horizon (Sharpe +0.79) and 63d horizon (Sharpe +0.58) are the strongest; 1d and 21d are negative — a bimodal reversal/drift profile rather than a monotone one.

Reproduce:

```
pip install -r requirements.txt
jupyter notebook notebooks/pipeline.ipynb # runs §0-§8
streamlit run app.py # dashboard
```